



OPEN In situ epoxidation of hybrid waste cooking oil and oleic acid via peracid mechanism

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The increasing generation of waste cooking oil (WCO) poses significant environmental challenges, making its valorization essential for sustainable waste management. This research investigates the in situ peracid method for epoxidizing a hybrid mixture of oleic acid and waste cooking oil. A novel approach is proposed by utilizing hybrid raw materials in the presence of natural zeolite as a catalyst to enhance epoxidation efficiency. The signal-to-noise (S/N) ratio analysis in Taguchi method showed that the optimum process parameters for production of epoxidized hybrid oleic acid and waste cooking oil to the response of relative conversion to oxirane (RCO) with determination of oxirane oxygen content (OOC) was maximum (50%) under following conditions: temperature of 50 °C, stirring at 100 rpm, and a molar ratio 1:1 with formic acid. After 100 iterations, the reaction rate constant based on optimized epoxidized hybrid oleic acid and waste cooking oil production was obtained as follows: $k_{11} = 13.45 \text{ mol} \cdot \text{L}^{-1} \cdot \text{min}^{-1}$, $k_{12} = 14.08 \text{ mol} \cdot \text{L}^{-1} \cdot \text{min}^{-1}$, $k_2 = 0.023 \text{ mol} \cdot \text{L}^{-1} \cdot \text{min}^{-1}$, and $k_3 = 0.025 \text{ mol} \cdot \text{L}^{-1} \cdot \text{min}^{-1}$. This discovery helps reduce waste, turns used cooking oil into a valuable commodity, and offers insight into reaction kinetics, a critical concept for industrial applications that is environmentally friendly.

Keywords Epoxidation, Biomass conversion, Kinetic model, Waste cooking oil

Global palm oil production is projected to reach 78–79 million metric tons in 2023/24, representing a 10% increase from the previous year¹. This steady production growth ensures a consistent and abundant palm oil supply, making it a viable and sustainable feedstock for industrial applications such as epoxidation. Malaysia is the second largest palm oil producer globally, providing approximately one-third of the world's palm oil supply. Palm oil represents a significant industry in Malaysia, contributing 2.4% to the country's total gross domestic product (GDP)². Malaysia's domestic palm oil consumption saw a rise to approximately 3.37 million metric tons in 2022. This figure is anticipated to climb due to population expansion and consumer favouritism for palm oil cooking oil³. However, improperly discarding waste cooking oil (WCO) can lead to notable environmental issues like clogging drains and sewers and contaminating water or soil. WCO is classified as hazardous waste due to these potential problems⁴. The oil is frequently poured into drains, causing harm to the ecological balance of rivers and seas⁵.

Globally, waste cooking oil (WCO) production surpasses 29 million tons annually, with improper disposal contributing to environmental pollution and health hazards⁶. Converting WCO into high-value products like epoxides supports sustainability while mitigating waste-related issues. Oleic acid, a highly reactive unsaturated fatty acid, enhances epoxidation efficiency and oxirane yield when combined with WCO. The growing demand for sustainable and environmentally friendly chemical processes necessitates the exploration of alternative catalysts that minimize ecological impact. While synthetic zeolites are widely recognized for their catalytic

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properties, their production involves complex and energy-intensive processes, leading to higher environmental and economic costs. In contrast, natural zeolites present a promising solution, as they are abundant, readily available, and require minimal processing⁷. This research addresses the challenge of utilizing WCO as a valuable resource through the epoxidation of hybrid oleic acid. Choosing natural zeolite as a heterogeneous catalyst aims to leverage its catalytic potential while reducing the ecological footprint associated with synthetic alternatives⁸. This study seeks to demonstrate that natural zeolites can provide an effective and sustainable pathway for the epoxidation process, ultimately contributing to waste valorization and the development of greener industrial practices.

Epoxidized vegetable oils present an attractive, cost-effective, renewable option for various industrial uses, as they exhibit similar properties to traditional petroleum-based epoxy thermosets⁹. The widespread interest in epoxy resins stems from the versatility of the epoxy group, which can undergo numerous chemical reactions and contribute to desirable properties in the final products¹⁰. The epoxy group, known as the oxirane ring, is highly reactive and readily undergoes ring-opening reactions. It can convert into polyols through alcoholysis in the presence of alcohols or thiols or hydrolysis when acid catalysts are used, and it also forms polyols through hydrogenation¹¹. The study focuses on optimizing the epoxidation process. Still, it does not explore its integration into circular economy frameworks or assess the market competitiveness of the final product in terms of cost and value-added benefits. The Taguchi method is a statistical optimization approach that enhances process efficiency by systematically analyzing multiple factors with minimal experimental trials. Using an orthogonal array design, it identifies key parameters influencing the reaction while minimizing variability. This study applied the Taguchi method to optimize epoxidation conditions, maximizing oxirane yield and minimizing side reactions. Its significance lies in improving process reliability, reducing resource consumption, and enhancing overall reaction performance¹².

Oleic acid, known for its high double-bond content, is widely used in epoxidation due to its efficient ability to form oxirane rings¹³. However, relying solely on oleic acid raises sustainability concerns, especially in large-scale applications, because of its high cost and limited availability. While it achieves a high oxirane content, exploring alternative or hybrid feedstocks is essential to improve the process's economic viability and environmental sustainability. The use of WCO as alternative feedstock has been explored to promote sustainability; however, challenges such as inconsistent feedstock composition, lower epoxidation efficiency, and undesired side reactions remain¹⁴. Additionally, most studies rely on strong acid catalysts, which contribute to equipment corrosion and require extensive purification steps¹⁵. A hybrid approach can overcome these limitations by combining oleic acid and WCO, enhancing process performance while promoting waste vaporization.

In contrast to traditional synthetic catalysts that involve complex and energy-intensive production, natural zeolites are abundant, readily available, and environmentally friendly¹⁶. This research presents a novel method for the epoxidation of hybrid oleic acid and WCO using natural zeolite as a heterogeneous catalyst. Combining hybrid oleic acid with waste cooking oil facilitates the production of high-value epoxidized products while addressing environmental concerns related to waste management. Ultimately, the research emphasizes natural zeolites as cost-effective and eco-friendly alternatives to conventional synthetic catalysts in industrial applications. Therefore, the aims of this research are (1) to optimise the process parameter of hybrid oleic acid and WCO with applied natural zeolite, and (2) to construct a kinetic model of epoxidation of hybrid oleic acid and WCO.

Methodology

Materials

The reagents employed in this study were Waste cooking palm oil from household waste, oleic acid (90%), hydrogen peroxide 50%, formic acid 94%, and natural zeolite (Clinoptilolite) as a catalyst purchased from Crystalize violet and QreC. Natural zeolite was used to promote the in situ formation of peracids and enhance the epoxidation rate by providing active sites for hydrogen peroxide activation and formic acid.

Experimental setup and procedure

The epoxidation procedure was performed in a fume hood to protect the experimenter from chemical exposure. 50 g of waste cooking oil with 50 g oleic acid mixture were combined in a beaker. The mixture was then water-bathed on a hot plate until it reached the desired temperature of 50–80 °C by using a thermometer. Subsequently, the mixture was stirred, and a constant solution of hydrogen peroxide, formic acid, and natural zeolite as a catalyst was added. Natural zeolite was used as a catalyst to replace conventional organic acids, offering higher stability, reusability, and reduced environmental impact while enhancing epoxidation efficiency. Natural zeolite sourced from Bukit Merah, Malaysia, is a microporous aluminosilicate mineral used as a heterogeneous catalyst in the epoxidation process due to its high surface area (BET surface area: 18.9 m²/g). Unlike conventional acid catalysts, natural zeolite is reusable, thermally stable, and environmentally friendly, making it a sustainable alternative. The mixture was maintained at a required temperature. The results of the titration of hydrobromic acid were recorded every 10-minute interval.

Experimental design

The various reaction conditions are summarized in Table 1 to assess the influence of different parameters on the epoxidation of hybrid oleic acid. The process was carried out in multiple trials to examine the effect of these conditions on the epoxidation rate, as shown in Table 1.).

Optimization using the Taguchi method

Optimization of the Taguchi Method process was carried out. The optimum result is analysed and predicted using the signal-to-noise ratios. The Taguchi method employs the signal-to-noise (S/N) ratio as a metric to

Reaction parameters	Range	Units
Reaction temperature	50–80	°C
Speed	100–400	rpm
Formic acid to hybrid oil	0.5–2.0	-

Table 1. Reaction parameters investigated for the overall reaction.

Parameters		Levels			
		1	2	3	4
A	Reaction temperature (°C)	50	60	70	80
B	Agitation speed (rpm)	100	200	300	400
C	Molar ratio of formic acid to hybrid oil	0.5	1	1.5	2

Table 2. Design of experiment, using three parameters and levels for the reaction optimization.

Experiment	Parameters and their level		
	Reaction temperature (°C)	Agitation speed (rpm)	Formic acid to oleic acid molar ratio
1	1	1	1
2	1	2	2
3	1	3	3
4	1	4	4
5	2	1	2
6	2	2	1
7	2	3	4
8	2	4	3
9	3	1	3
10	3	2	4
11	3	3	1
12	3	4	2
13	4	1	4
14	4	2	3
15	4	3	2
16	4	4	1

Table 3. Orthogonal array used to design experiment for the reaction parameter.

evaluate the effectiveness of a process. By leveraging the S/N ratio, one can assess the impact of different design factors on process performance and pinpoint areas for enhancement¹⁷. This approach helps reduce variability and enhance reliability, ultimately leading to notable gains in process efficiency and quality¹⁸. Additionally, since the Taguchi method follows a structured and systematic approach, it is relatively straightforward to implement and sustain, making it a highly appealing choice for process optimization. This approach optimizes reaction parameters with minimal experimental trials using orthogonal arrays¹⁹. All these variables were evaluated at different levels, as shown in Table 2.

In this experiment, three parameters were selected: temperature, agitation speed, and the ratio of formic acid to hybrid oil. High oxirane oxygen conversion and relative conversion to oxirane (RCO) are particularly significant.

The orthogonal arrays used have three parameters at three levels. Crossing these, as shown in Table 3, enables the investigation of factor diversity.

Fourier transform infrared spectroscopy

Infrared spectroscopy is one of the most widely used techniques for chemical analysis and is readily available in many laboratories, with Fourier transform infrared spectroscopy (FT-IR)²⁰. This method provides insights into molecular vibrations and helps identify functional groups, making it an effective tool for determining chemical components, molecules, and molecular structures²¹. The resulting IR spectrum, which displays a series of absorbance bands (either as absorbance or transmittance intensity versus frequency or wave number), reveals structural and bonding information within a molecule, as different wavelengths correspond to distinct functional groups.

Determination of oxirane content (OOC)

The titration method was used to determine the oxirane oxygen content (OOC)²². A number of procedures were taken to identify a sample's OOC. A sample size of 3 g was needed for accurate findings. This aided in depolymerisation, which split bigger molecules into smaller ones. One drop of crystalize violet was used as an indicator. Following the completion of this reaction, the sample was titrated using hydrobromic acid. The OOC of the sample was determined by measuring the volume of bromide required for titration when the purple colour changed into yellowish green²³. The unsaturated fatty acids had been successfully transformed into epoxidised fatty acids, as evidenced by the oxirane ring in Eqs. 4–6.

$$RCO = \frac{OOC_{\text{experimental}}}{OOC_{\text{theoretical}}} \times 100 \quad (1)$$

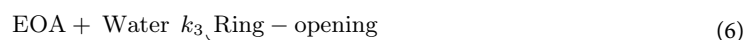
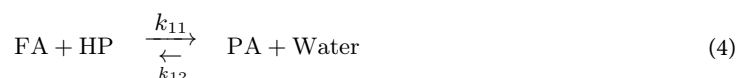
$$OOC_{\text{theoretical}} = \left\{ \left(\frac{X_0}{A_i} \right) / \left[100 + \left(\frac{X_0}{2A_i} \right) (A_o) \right] \right\} \times A_o \times 100 \quad (2)$$

$$OOC_{\text{experimental}} = 1.6 \times N \times \frac{(V-B)}{W} \quad (3)$$

Here, A_i is the molar mass of iodine, A_o is the molar mass of oxygen, X_0 is the initial iodine value, N is the normality of HBr, V is the volume of HBr solution used for titration, and W is the weight of the sample.

Kinetic modelling of the in situ epoxidation palm stearin

The kinetic data were modeled based on the following assumptions: (1) the epoxidation occurs as a single-phase process, eliminating the need for distribution constants between aqueous and oil phases; (2) phase volumes remain constant throughout the reaction; (3) all reactions are homogeneous; (4) heat transfer effects are negligible; and (5) reactions occur away from the phase interface. It involves the in situ formation of performic acid (PA) (Eq. 4) and the formation of EOA (Eq. 5), with ring opening shown in Eq. 6.



Where FA, HP, OA, and EOA are formic acid, hydrogen peroxide, hybrid oleic acid, and epoxidized hybrid oleic acid. Kinetic modeling involves two main steps: (1) numerically solving the rate equations and (2) calculating the error between simulated and experimental data. The MATLAB ode45 solver, which applies the fourth-order Runge–Kutta method, was used to solve the system of rate equations. The rate equations were modelled, resulting in simultaneous differential Eqs. 7–13.

$$\frac{d[FA]}{dt} = -k_{11} [FA] [HP] + k_{12} [PA] [\text{Water}] + k_2 [PA] [OA] \quad (7)$$

$$\frac{d[HP]}{dt} = -k_{11} [FA] [HP] + k_{12} [PA] [\text{Water}] \quad (8)$$

$$\frac{d[PA]}{dt} = +k_{11} [FA] [HP] - k_{12} [PA] [\text{Water}] - k_2 [PA] [OA] \quad (9)$$

$$\frac{d[\text{Water}]}{dt} = +k_{11} [FA] [HP] - k_{12} [PA] [\text{Water}] - k_3 [EOA] [\text{Water}] \quad (10)$$

$$\frac{d[OA]}{dt} = -k_2 [PA] [OA] \quad (11)$$

$$\frac{d[EOA]}{dt} = +k_2 [PA] [OA] - k_3 [EOA] [\text{Water}] \quad (12)$$

$$\frac{d[\text{Ring opening}]}{dt} = +k_3 [EOA] [\text{Water}] \quad (13)$$

The experimental procedure involves feedstock preparation (WCO and oleic acid), catalyst addition (natural zeolite), and in situ peracid formation (hydrogen peroxide and formic acid). The epoxidation reaction is conducted under controlled conditions, followed by kinetic mode, as shown in Fig. 1.

Result and discussion

Fatty acid composition

The composition of fatty acids plays a vital role in the epoxidation process, as the reaction occurs specifically at the double bonds found in unsaturated fatty acids. Table 4 highlights the fatty acid composition of pure oleic acid, waste cooking oil (WCO), and their hybrid blend analyzed using Gas Chromatography-Flame Ionization

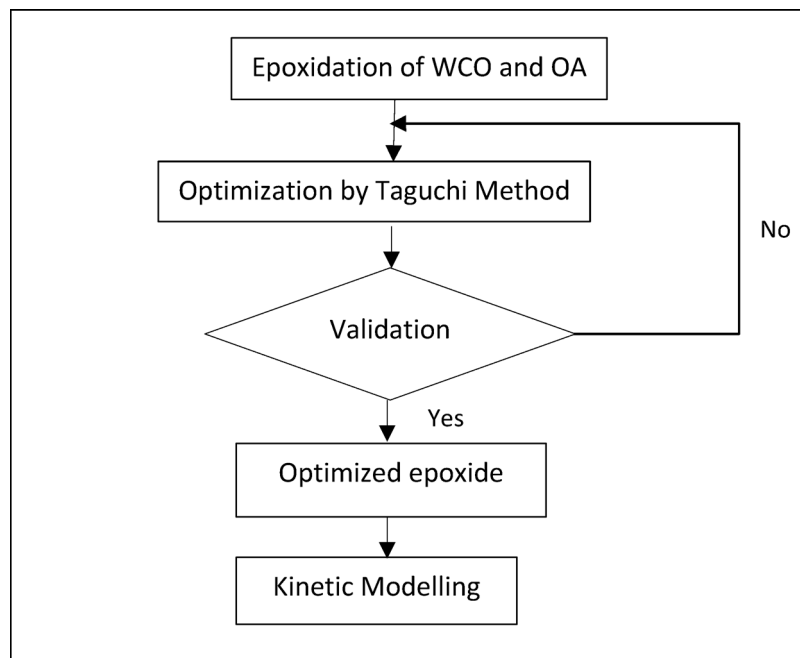


Fig. 1. Flowchart of overall methodologies.

Type of oil	Unsaturated fatty acid (%)	Saturated fatty acid (%)
Pure oleic acid	85	15
WCO	51	49
Hybrid oleic acid + WCO	62	38

Table 4. Fatty acid composition of hybrid oleic acid and WCO.

Detection (GC-FID). Pure oleic acid, comprising 85% unsaturated fatty acids, demonstrates the greatest potential for epoxidation due to its high double-bond content. On the other hand, WCO has a lower percentage of unsaturated fatty acids (51%), which limits its reactivity. The hybrid mixture, containing 62% unsaturated fatty acids, represents a balanced approach, combining the reactivity of oleic acid with the sustainability benefits of WCO. This blend optimizes feedstock utilization while maintaining sufficient double-bond availability for effective epoxidation.

Taguchi analysis

The Taguchi method was applied to predict grinding surface roughness by organizing data through orthogonal experiments, as shown in Table 5. It treats error factors as sources of interference that cause fluctuations in prediction outcomes and identifies the primary and secondary factors influencing surface roughness using S/N ratio analysis and ANOVA. The sequence of experiments, based on a combination of parameters and levels, is structured using an orthogonal array, which defines the number of trials needed to ensure that each factor's levels are equally tested. Taguchi introduced an orthogonal array to assess the influence of individual process parameters. L16 array, consisting of 16 trials, tests four levels across up to five experimental factors. Using this specific Taguchi orthogonal array, only 16 runs are necessary to optimize four levels across four factors, significantly reducing the number of experimental trials required for optimization, as shown in Table.

Main effect plot for signal to noise ratios

Figure 2 shows the ideal parameters for synthesizing waste cooking oil-oleic acid: a reaction temperature of 50 °C for optimal epoxidation yield, maintaining temperatures below 100 °C to prevent explosions from hydrogen peroxide, a speed of 100 rpm to avoid overmixing, and a 1:1 molar ratio of formic acid to oil for maximum oxirane conversion. The primary effect plot illustrates the impact of each component on performance based on signal-to-noise (SN) ratios²⁴. It rapidly evaluates each component's impact by displaying the average SN ratio at various levels for each factor.

The Taguchi method is a novel optimization technique that uses the design of experiments to find ideal parameter sets. It efficiently arranges variable parameters at different levels through orthogonal arrays, minimizing the number of experiments required. Throughout the procedure, the signal-to-noise ratio (S/N) is assessed for each parameter level; the best level is indicated by the maximum S/N value, as shown in Table 6. The analysis indicates that temperature significantly affects the epoxidation process, followed by stirring speed, while the

Temperature (°C)	Speed (rpm)	Formic acid (g)	RCO
50	100	16	45.35
50	200	8	27.21
50	300	5.3	27.21
50	400	4	27.21
60	100	16	27.21
60	200	8	18.14
60	300	5.3	18.14
60	400	4	18.14
70	100	16	18.14
70	200	8	18.14
70	300	5.3	18.14
70	400	4	18.14
80	100	16	18.14
80	200	8	18.14
80	300	5.3	18.14
80	400	4	18.14

Table 5. Parameters value of epoxidation and RCO.

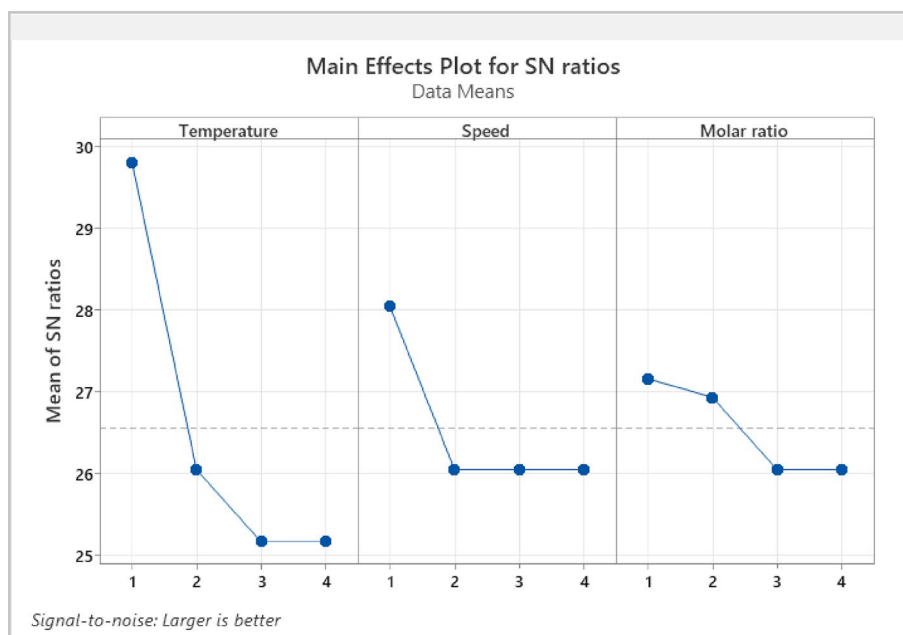


Fig. 2. Main effects plot for SN ratios.

Level	Temperature	Speed	Molar ratio
1	29.81	28.04	27.16
2	26.05	26.05	26.94
3	25.17	26.05	26.05
4	25.17	26.05	26.05
Delta	4.63	1.99	1.11
Rank	1	2	3

Table 6. Response for signal-to-noise ratios.

Source	DF	Adj SS	Adj MS	F-value	P-value
Temperature	3	509.2	169.73	6.60	0.007
Error	12	308.6	25.72		
Total	15	817.8			

Table 7. ANOVA table for RCO versus reaction temperature.

Source	DF	Adj SS	Adj MS	F-value	P-value
Temperature	3	138.9	46.29	0.82	0.508
Error	12	678.9	56.58		
Total	15	817.8			

Table 8. ANOVA table for RCO versus reaction speed.

Source	DF	Adj SS	Adj MS	F-Value	P-Value
Temperature	3	56.58	18.86	0.30	0.827
Error	12	761.20	63.43		
Total	15	817.8			

Table 9. ANOVA table for RCO versus reaction of formic acid.

molar ratio has the least impact. Temperature significantly impacts epoxidation by accelerating reaction kinetics and peracid formation, while excessive heat may promote side reactions. Stirring speed enhances mass transfer, ensuring efficient reactant mixing, catalyst interaction, and higher epoxide yield. In contrast, the molar ratio has minimal effect, as excess reactants beyond an optimal level do not significantly improve conversion. Optimizing these parameters can enhance the efficiency of the epoxidation of hybrid oleic acid and waste cooking oil.

One way analysis of variance (ANOVA)

One-way ANOVA is a method for comparing averages across different groups or conditions²⁵. In the context of RCO, one-way ANOVA determines whether variables such as temperature, agitation speed, and molar ratio significantly influence RCO. To do this, several experiment variations are made by varying the temperature, the agitation speed, and the molar ratio. The one-way ANOVA results are customarily presented as F-statistics coupled with a p-value where the p-value expresses the probability of the testing status. High p-values indicate that factors under consideration (temperature, agitation speed, and molar ratio) cause less or no changes to the disparities in RCO. Suppose the p-value obtained is low compared to the a priori-determined level, usually chosen at a 5% level. In that case, the differences are significant, so RCO is most probably affected by these factors, and the differences likely did not occur by chance.

Following one-way ANOVA also provides guidance on whether temperature, agitation speed, and molar ratio beliefs that adjustments in these variables will improve RCO cut-off and what should be the optimal conditions. In the present study, the null hypothesis postulates that the different RCO mean values taken at different time points are assumed to be equal.

Table 7 illustrates the interpretation of the data with a p-value of 0.007 and temperature values of 50, 60, 70, and 80. The significance level (α) set at 0.05 possesses adequate evidence to reject the null hypothesis, as the p-value (0.007) is less than the significance level ($\alpha = 0.05$). This indicates that the data may be used to deduce that the mean RCOs at the various temperature points differ significantly. Therefore, data variability or random chance cannot account for the observed changes in RCO at 50, 60, 70, and 80 °C. According to this investigation, there is a lot of evidence that suggests temperature affects RCO.

With speeds of 100, 200, 300, and 400, the analysis yielded a p-value of 0.508, indicating insufficient evidence to reject the null hypothesis. This shows that the observed differences in RCO at different speeds could be caused by random chance or intrinsic data variability. Table 8 shows that the speed did not have a statistically significant effect on RCO within the confines of this study.

A computed p-value of 0.827 was obtained from the analysis using molar ratios of 0.5, 1.0, 1.5, and 2.0. There was not enough data to reject the null hypothesis because the obtained p-value was higher than the significance level. This implies that data variability or random chance may account for changes in RCO at different molar ratios. The results indicate that the molar ratio did not significantly impact RCO in this investigation, as shown in Table 9.

Epoxidation of hybrid oleic acid and waste cooking oil

Figure 3, depicting the relative conversion to oxirane (%) over time during the epoxidation of hybrid oleic acid and waste cooking oil, provides critical insights into the reaction kinetics of this process. The data demonstrate a progressive increase in conversion, reaching a maximum of approximately 50% at around 40 min, which

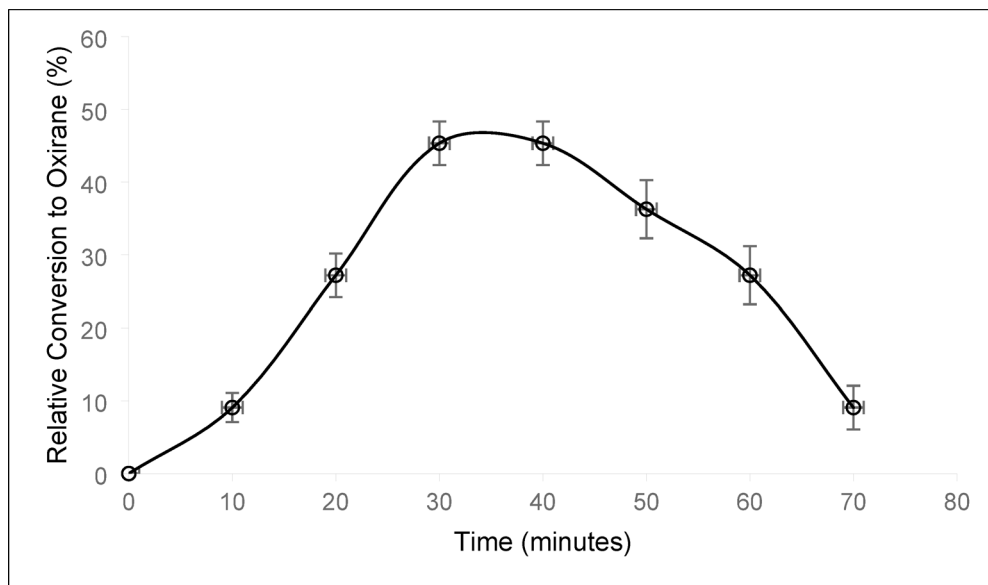


Fig. 3. The epoxidation of hybrid oleic acid and waste cooking oil.

Reaction rate constant	Value (mol/L·min)
k_{11}	13.45
k_{12}	14.08
k_2	0.023
k_3	0.025

Table 10. Reaction rate constants estimated from Simulations. Correlation coefficient, $r = 0.90$.

indicates effective interaction dynamics between the hybrid oleic acid, WCO, and the oxidizing agent employed. This peak conversion suggests that the epoxidation conditions are optimized within this timeframe, facilitating the transformation of a significant proportion of unsaturated fatty acid bonds into epoxide groups, thereby enhancing the potential for generating high-value products²⁶. Previous studies indicate that catalyst loading and hydrogen peroxide concentration have minimal effect on the maximum achievable RCO but significantly influence the time required to reach it^{1,16}. Higher peroxide levels and catalyst loadings accelerate the epoxidation rate, enabling faster attainment of peak oxirane content, though the ultimate RCO remains essentially unchanged.

Based on previous research findings, the impact of catalyst type and hydrogen peroxide amount was not explicitly analyzed as these parameters were kept constant²⁷. However, the subsequent decline in conversion after 50 min may indicate that the reaction is approaching equilibrium, or it could be attributed to side reactions, product degradation, or depletion of reactants. The reagent concentration and amounts significantly influenced the formation of side products in the epoxidation process. Excess peracid led to the hydrolysis of oxirane rings, resulting in diols as a major byproduct. Additionally, over-oxidation at high oxidantThe error bars presented in the graph reflect the variability of the data, with relatively small bars indicating a high degree of reliability and reproducibility in the experimental results. This study underscores the potential of utilizing waste cooking oil as a valuable resource for producing epoxidized chemicals, promoting sustainable chemical industry practices.

Numerical kinetic modelling of epoxidation hybrid oil

Kinetic data provide a critical foundation for process scale-up, offering predictive insight essential for industrial applications where both high reaction efficiency and long-term product stability are key requirements²⁸. Table 10 provides the estimated reaction rate constants derived from the kinetic modeling of the epoxidation process. The parameters k_{11} and k_{12} correspond to the formation of peracid and the subsequent epoxidation of unsaturated bonds, with their relatively large values (13.45 and 14.08 mol/L·min) indicating rapid reaction progress under the experimental conditions. Conversely, the smaller constants k_2 and k_3 (0.023 and 0.025 mol/L·min) reflect slower secondary processes, likely involving oxirane ring-opening and peroxide breakdown. These slower reactions are more noticeable during extended reaction periods and contribute to the reduction in oxirane content over time. The correlation coefficient of 0.90 suggests that the model provides a good fit to the experimental data, capturing the main kinetic behavior effectively, although further refinement may improve accuracy for side reactions and catalyst-related effects.

The correlation coefficient was 0.90, showing a reasonable agreement between simulation and experimental data, as shown in Fig. 4. The mismatch between the kinetic model and experimental data, particularly at

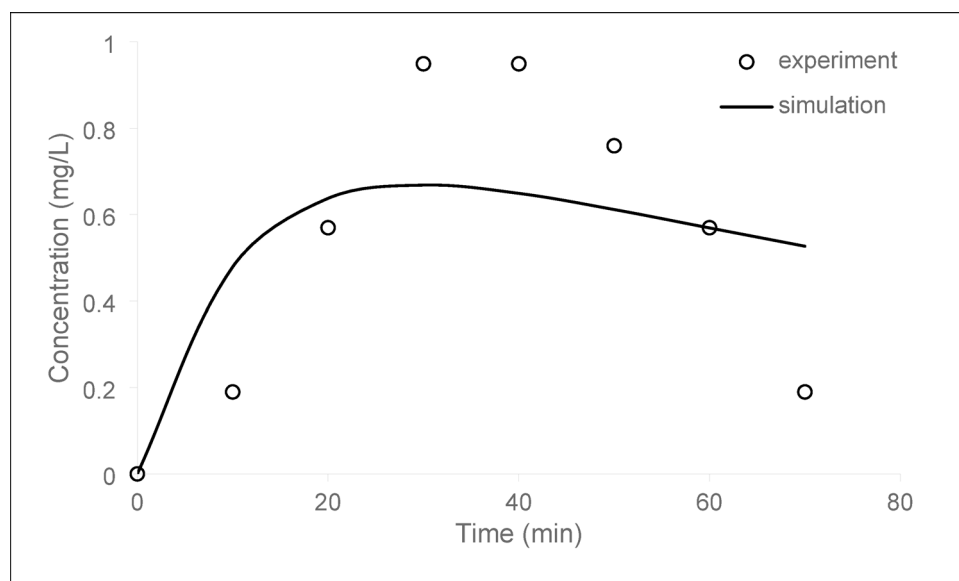


Fig. 4. Comparison of the oxirane content between simulation and experiment.

extended reaction times, is attributed to parasitic effects and non-ideal behaviors not fully captured by the model. Although the model assumes a homogeneous single-phase system, localized heterogeneities likely exist due to partial immiscibility between hydrogen peroxide and the oil phase, leading to concentration gradients and deviations in reaction rates. Additionally, side reactions such as hydrolysis and alcoholysis of epoxide rings and peroxide decomposition become significant over time, reducing oxirane content and increasing hydroxyl values beyond model predictions. Although the model does not account for catalyst deactivation, prolonged reaction times could lead to gradual fouling of the zeolite's active sites due to adsorption of organic by-products, structural changes, or leaching of minor elements. This deactivation would reduce the effective catalytic activity over time, causing slower-than-predicted epoxidation rates.

In contrast, the experimental process cannot account for any vaporized gases. Therefore, the epoxidation and oxirane cleavage times are expected to differ between simulation and experiment. It is not possible to match the maximum oxirane content from the simulation with the experimental results, as the simulation assumes a one-way reaction.

Conclusion

This research successfully demonstrates the feasibility of using the in situ peracid method to produce epoxidized hybrid oleic acid and waste cooking oil (WCO), offering a novel and sustainable approach to WCO vaporization. This study introduces an environmentally friendly process that can convert waste into valuable epoxidised products by employing hybrid raw materials oleic acid and WCO and utilizing natural zeolite as a catalyst. The optimization of reaction parameters using the Taguchi method revealed that a temperature of 50 °C, stirring at 100 rpm, and a 1:1 molar ratio with formic acid are the ideal conditions for synthesizing epoxidized oleic acid from WCO. The highest epoxidized hybrid oleic acid and WCO yield (50%) was obtained under these optimized reaction conditions. ANOVA analysis confirmed that the temperature was the most influential factor in the epoxidation process. Finally, the process model was developed using the Runge-Kutta 4th-order numerical integration method to determine the kinetic model. The Genetic Algorithm was applied to identify the model that best fits the experimental data. These findings provide valuable insights into reaction kinetics, which are crucial for scaling up the process for industrial applications. Overall, this work contributes to sustainable waste management by transforming WCO into a high-value product while minimizing environmental impact.

Data availability

The data that support the findings of this study are available from the corresponding author upon reasonable request.

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Author contributions

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Declarations

Competing interests

The authors declare no competing interests.

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